

COMPANION
to

ChatGPT Teaches TikZ

Chapter 16: More Geometric Constructions (I)

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16.1 What This Companion Adds

Chapter 16 constructed the incircle of a triangle. This Companion makes the construction reusable: it calculates the incenter from the three side lengths, projects the incenter onto a side, and uses that perpendicular distance as the radius.

Design

The incenter and incircle radius are consequences of the triangle. Let the source encode those dependencies instead of storing approximations.

16.2 An Incircle Vocabulary

Form	Purpose
<code>\$(B)-(C)\$</code>	Forms the vector difference of two coordinates.
<code>\p1</code>	Stores a temporary point or vector.
<code>\n1</code>	Stores a temporary length.
<code>vecLen(x,y)</code>	Calculates a vector length.
<code>barycentric cs:</code>	Defines a weighted combination of vertices.
<code>\$(A)!(I)!(B)\$</code>	Projects I onto the line through A and B.
<code>let ... in</code>	Performs local calculations inside a path command.
<code>circle (\n1)</code>	Uses a calculated length as a radius.

The `calc` library supplies the coordinate calculations.

16.3 The Mathematical Dependency Chain

For triangle ABC , write

$$a = |BC|, \quad b = |CA|, \quad c = |AB|.$$

The incenter has barycentric weights $a : b : c$. Once the incenter I is known, project it onto a side at D . The distance ID is the incircle radius.

16.4 Computing the Side Lengths

```
\path
let
  \p1 = ($ (B) - (C) $),
  \p2 = ($ (C) - (A) $),
  \p3 = ($ (A) - (B) $),
  \n1 = {vecLen(\x1,\y1)},
  \n2 = {vecLen(\x2,\y2)},
  \n3 = {vecLen(\x3,\y3)}
in
coordinate (I) at
(barycentric cs:A=\n1,B=\n2,C=\n3);
```

The weight at each vertex is the length of the opposite side.

Common Mistake

Do not attach $|AB|$ to vertex A . Vertex A receives the length of the opposite side BC .

16.5 Marking the Incenter

After the calculation, I behaves like any named coordinate.

```
\filldraw (I) circle (2pt) node[right] {$I$};
\draw[densely dashed] (A)--(I) (B)--(I);
```

Only two angle bisectors are needed to locate their common intersection. The third may be drawn as a check.

16.6 Projecting onto a Side

Let D be the perpendicular projection of I onto the line through A and B .

```
\coordinate (D) at ($ (A)!(I)!(B)$);  
\draw[thin] (I)--(D);
```

The middle coordinate is the point being projected; the first and last points define the target line.

16.7 Using the Perpendicular Distance

```
\draw[thick]  
let  
  \p1 = (I),  
  \p2 = (D),  
  \n1 = {veclen(\x2-\x1,\y2-\y1)}  
in  
  (I) circle (\n1);
```

The radius changes automatically when the triangle changes.

16.8 Why the Circle Touches Every Side

A point on an angle bisector is equally distant from the two sides of that angle. Because I lies on two interior angle bisectors, it is equally distant from all three sides. Therefore a circle centered at I with radius ID is tangent to all three sides.

16.9 Worked Project: A Reusable Incircle

16.9.1 Stage 1: Name the Triangle

```
\coordinate (A) at (-2,0);  
\coordinate (B) at (3,0);  
\coordinate (C) at (.5,3);
```

16.9.2 Stage 2: Calculate the Center and Radius Point

```
% Calculate I from opposite side lengths.  
% Then project I onto AB.  
\coordinate (D) at ($(A)!(I)!(B)$);
```

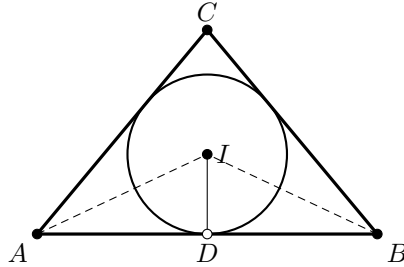
16.9.3 The Complete Picture

```
\begin{tikzpicture}
  \coordinate (A) at (-2,0);
  \coordinate (B) at (3,0);
  \coordinate (C) at (.5,3);

  \path let
    \p1=($(B)-(C)$), \p2=($(C)-(A)$),
    \p3=($(A)-(B)$),
    \n1={veclen(\x1,\y1)},
    \n2={veclen(\x2,\y2)},
    \n3={veclen(\x3,\y3)}
  in coordinate (I) at
    (barycentric cs:A=\n1,B=\n2,C=\n3);

  \coordinate (D) at ($(A)!(I)!(B)$);
  \draw[very thick] (A)--(B)--(C)--cycle;
  \draw[densely dashed] (A)--(I) (B)--(I);
  \draw[thin] (I)--(D);
  \draw[thick]
    let \p1=(I), \p2=(D),
        \n1={veclen(\x2-\x1,\y2-\y1)}
    in (I) circle (\n1);
  \filldraw (A) circle (2pt) node[below left] {$A$};
  \filldraw (B) circle (2pt) node[below right] {$B$};
  \filldraw (C) circle (2pt) node[above] {$C$};
  \filldraw (I) circle (2pt) node[right] {$I$};
  \filldraw[fill=white] (D) circle (2pt) node[below] {$D$};
\end{tikzpicture}
```

See



Design

Only the three vertices are independent. The side lengths determine I , the projection determines D , and the segment ID determines the circle.

16.10 Comparing the Two Circle Constructions

The circumcenter is obtained from perpendicular bisectors of sides and is equidistant from the vertices. The incenter is obtained from angle bisectors and is equidistant from the sides. Similar TikZ workflows encode different geometric facts.

16.11 Debugging Workshop

16.11.1 The Incenter Leaves the Triangle

Check the barycentric weights. For a nondegenerate triangle, positive opposite side lengths place the incenter inside the triangle.

16.11.2 The Radius Is Not Perpendicular

Use the projection syntax with I in the middle and the two endpoints of one side outside.

16.11.3 The Circle Misses a Side

Calculate the radius from I to the projected point D ; do not type an approximate number.

16.11.4 The Construction Lines Dominate

Use a lighter dashed style for the angle bisectors and a heavier style for the triangle and incircle.

16.12 Exercises

1. Move vertex C and verify that the incircle follows.
2. Add the third angle bisector.
3. Project I onto side AC instead of AB .
4. Mark the three tangency points.
5. Define one shared style for all auxiliary lines.
6. Compare the three calculated perpendicular distances.
7. Hide the angle bisectors while retaining the incircle.
8. Combine the incircle and circumcircle in one triangle.

16.13 Solutions

16.13.1 Exercises 1–4

Exercise 1 changes only the coordinate of C.

```
\draw[densely dashed] (C)--(I);
```

```
\coordinate (E) at ($(A)!(I)!(C)$);
```

For Exercise 4, project I onto each of the three side lines and mark the resulting coordinates.

16.13.2 Exercises 5–7

```
\tikzset{aux/.style={densely dashed,gray}}
```

Exercise 6 measures the distances from I to the three projection points; they agree. Exercise 7 removes only the drawing commands for the angle bisectors.

For Exercise 8, keep separate names for the circumcenter and incenter and use different line styles for their circles.

16.14 ChatGPT Workshop

Ask ChatGPT to check the geometry before simplifying the TikZ source.

1. Verify that each barycentric weight is attached to the opposite vertex.
2. Replace this approximate incenter with a calculation from side lengths.
3. Diagnose why this projection point does not lie on side AB .
4. Compute the incircle radius without typing a numerical length.
5. Compare this incircle construction with the circumcircle construction.

16.15 Final Checklist

Before continuing to Chapter 17, check that you can

1. compute the three opposite side lengths;
2. use barycentric coordinates to define the incenter;
3. project a point onto a line;
4. calculate a radius from two dependent points;
5. explain why the circle is tangent to all three sides;
6. distinguish given, auxiliary, and final geometry; and
7. change the triangle without rebuilding the construction.